Chapter 32 Back Matter Check

Contents

32 Supersymmetry Algebras in Higher Dimensions

Problems

1. Classify the massless particle multiplets for each allowed kind of supersymmetry in six spacetime dimensions, when all central charges vanish.

2. Suppose that it were possible to have massless particles for all spins up to $j = 3$, but no higher. Taking into account the fact that massless particles of spin 2 do exist, what is the maximum spacetime dimension in which supersymmetry is possible? What is the maximum number of supersymmetry generators for each allowed value of the spacetime dimensionality?

3. Consider types IIA and IIB supersymmetry in ten spacetime dimensions. Assume that only scalar central charges appear in the extended supersymmetry anticommutation relations. Find a lower bound on particle masses in terms of these central charges. Describe the 'BPS' massive particle multiplets allowed for particles whose masses are at this lower bound.

4. List the possible independent scalar and/or tensor central charges for $N = 1$ supersymmetry in nine spacetime dimensions.

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